Ahmad Choudhary

+1 (217)-220-6093 | asc113@duke.edu | github.com/2Ahmad5 | linkedin.com/in/ahmad-choudhary | ahmadchoudhary.dev

EDUCATION

Duke University *Durham, NC*

B.S. in Computer Science (Conc. in AI/ML); Minor in Physics & Math | GPA: 3.84 May 2027

Relevant Courses: Data Structures & Algorithms, Machine Learning, Deep Learning, Algorithmic Trading, Computer Architecture, Multivariable Calculus, Linear Algebra, Computer Vision, Databases, Natural Language Processing, Systems Programming

EXPERIENCE

Machine Learning Engineer Intern - Cruise (Autonomous Vehicles) *May 2026 – Aug 2026*

- Built SPARK, an **ML infrastructure** tool using **Pyroscope telemetry** to profile CPU waits, mutex contention, blocks, and memory allocation post-training and broad ML workflows.
- Designed an **AI-driven hotspot pipeline** that reduced regressions in ML training by ~**40%** through automated bottleneck detection.
- Implemented automated PR generation for targeted fixes, reducing contention/allocation overhead and improving workload performance by ~**12–18%.**

Machine Learning Researcher - Fain Lab Duke *Apr 2026 – Present*

- Conducting research on LLM safety using **Inverse reinforcement learning** to improve reward interpretability and model alignment.
- Investigating **preference-learning** methods by evaluating whether IRL models outperform standard Bradley–Terry algorithms.
- Developed evaluation pipelines to assess robustness of reward models against unfiltered LLM-generated preference outputs.

Software Engineering Intern - Casca (YC S23) *May 2025 – Feb 2026*

- Operated in a **forward-deployed model** with major banks (**Huntington, Live Oak, etc**.), deploying loan automation that cut SBA processing times by **10×**.
- Built **AI agents** that generated banker UIs directly from production schemas, reducing engineering requests by **60%**; implemented with **PostgreSQL, Prisma, and tRPC** for type-safe automation.
- Helped scale ARR from ~**$80K → $7M+** and supported fundraising of a **$29M Series A** through direct engineering contributions.
- Strengthened financial data security by optimizing **Plaid** integrations, improving **AWS ECS/S3** infrastructure, and hardening transaction handling.

Software Engineering Intern - Duke Fuqua School of Business *Sept 2024 – May 2025*

- Built a responsive frontend using **Svelte** and **TypeScript**, creating a seamless UI and enhancing code scalability
- Developed a robust backend with **Flask API** and Python to process real-time stock data, delivering accurate investment insights
- Deployed the platform using cloud hosting to ensure scalability and accessibility for **100+ Finance graduate** students

PROJECTS

Shahzad o3 | *NLP, LoRA, RL, Pytorch*

- Fine-tuned an **8B LLM** on Tinker using **LoRA, SFT, and RL** to improve instruction-following, reasoning, and code generation.
- Optimized multi-domain datasets with targeted sampling to improve performance and overall model generalization.
- Implemented **DPO** and **GRPO** reinforcement learning pipelines, increasing math accuracy from **44%** to **78.0%** across evaluations.

Explorer | *Agentic AI, Swift, Cerebras*

- Built an AI-first Finder/File Explorer with Swift for backend and frontend, optimized for low-latency **RAG** operations
- Implemented **agentic AI** for semantic file search, contextual understanding, and autonomous task execution across files and folders
- Integrated Google Docs and Gmail APIs to enable in-app document creation, saving, and rapid email workflows from the explorer

ML Games - Pac-man and Robotics | *Reinforcement Learning, Cross-Validation, Random Forest Classifier, Python*

- Applied **Reinforcement Learning** with a custom heuristic and robust score function to Pacman in order to achieve **90% win rate**.
- Leveraged Stat-botics API data to achieve an **85% success rate** in predicting outcomes for **6,000** matches across **2,500** teams
- Coded cross-validation for feature importance with Random Forest Classifiers and ran binary classification models, boosting data training accuracy by **20%** and refining prediction reliability for competition scenarios

TruthGuard | *React, Perplexity API, Flask, JavaScript, Vite*

- Created a Chrome extension for **real-time fact-checking**, analyzing **articles and videos** the user is viewing for accuracy.
- Utilized **Perplexity AI** API in order to get real time truths and reliabilities for information backed by real citations and evidence.
- Achieved **1st place at HackDuke**, leading frontend, backend, prompt engineering, and server integration, optimizing time.

Deep Learning Projects | *Pytorch, Tensorflow, Numpy, CNN, MLP*

- Investigating confounding variables in deep causal estimators such as **TARNet** to improve treatment effect estimation accuracy.
- Trained a custom CNN architecture for multi-class image classification, achieving **70% accuracy** on ImageNet dataset.
- Built a custom **Region Proposal Network** on top of the CNN backbone, increasing image search speed and detection efficiency.

SKILLS

Languages: Java, Python, C, C++, JavaScript, TypeScript, Bash, HTML/CSS, SQL, MIPS Assembly, C#, XML
Frameworks: Svelte, React, Node js, PyTorch, Flask, FastAPI, React-Native, WordPress, TensorFlow, TRPC, Prisma
Tools: Matplotlib, AWS, CAD, PostgreSQL, Docker, Vercel, Tailwind, Scikit Learn, Qiskit, Qt Widget, GitHub, FEBio, MongoDB